

Introduction

The varroa mite (*Varroa destructor* Korean haplotype) is a destructive pest of the European honey bee that spread worldwide during the late 20th century, killing unprotected beehives. Globally, only the Australian honey bee industry remains free of the pest.

In 2010 the Australian Government, through the then Department of Agriculture, Fisheries and Forestry, led a government–industry committee that consulted nationally with state and territory governments, honey bee and crop industries and the scientific community, and prepared a strategy to support the continuity of honey bee businesses, and the crop industries they support, in the event of the varroa mite becoming established in Australia. The strategy is available at www.agriculture.gov.au/honeybees.

The varroa mite is highly likely to establish in Australia in the future—the balance of expert and scientific opinion is that this is inevitable. If it does establish, it will transform our beekeeping and crop industries. The strategy recommends 10 actions to strengthen the capacity and preparedness of Australia’s honey bee and crop industries.

The strategy concludes that more research and development needs to be done now to help keep Australia free from varroa for as long as possible and to support Australian industries with the information, skills and tools they’ll need if the varroa mite establishes here.

The mite has transformed beekeeping industries in other countries where it must be controlled by chemical and other treatments. Its presence has also forced some overseas farmers to change the way they pollinate some crops. The effective eradication of wild European bees, which pollinate crops for free, has compelled them to make greater use of commercial pollination services, increasing their costs and sometimes leading to shortages in the number of hives available for crop pollination.

While Australia’s freedom from varroa mite is unlikely to last, it is important we make every effort to retain our varroa-free status for as long as possible. Despite best efforts, it is highly likely that varroa will establish here, affecting 1700 commercial honey bee businesses, 20 000 crop industry businesses and 10 000 hobby beekeepers.

Australian industries will face unique challenges in coping with the varroa mite. In particular, the nomadic nature of our honey bee industry, coupled with a high labour requirement and high labour costs, will make it harder and more costly to manage the pest.

Australian crop industries are likely to face higher pollination rental fees than those in other countries due to higher costs for bee keepers and greater competition for available pollination services.

The Varroa Continuity Strategy Management Committee¹ of industry, government and scientific representatives has identified four research and development priorities.

Key points

- We need to prepare for varroa mite— research and development is a vital part of a national strategy to protect our beekeeping and crop industries.
- Varroa mite is a destructive pest that kills unprotected beehives—it's highly likely the mite will establish in Australia in the future.
- Australian scientists and organisations involved in honey bee, varroa mite and crop pollination research should be guided by this Statement of research and development priorities as they develop research proposals.

The Department of Agriculture prepared this document in collaboration with:

- Australian Honey Bee Industry Council
- Almond Board of Australia
- CSIRO
- Department of Agriculture and Food Western Australia
- Horticulture Australia Limited
- New South Wales Department of Primary Industries
- Plant Health Australia
- Rural Industries Research and Development Corporation
- Victorian Department of Environment and Primary Industries

It is also endorsed by the Varroa Continuity Strategy Management Committee.

Research and development priorities

- Improving the efficiency and effectiveness of crop pollination under Australian conditions

In the absence of the pollination services provided by wild European honey bees, crop industries dependent on honey bee pollination (particularly higher value horticultural crops) will increasingly rely on beekeeper-provided commercial pollination services. This increased demand is likely to cause a significant rise in the price of pollination services. To support the adoption of paid pollination services and reduce the cost and potential yield effects on crop industries, research and development is needed to:

- quantify the contribution of wild honey bees and native insects to crop pollination
- improve the efficiency of crop pollination by managed honey bees (more pollination by fewer honey bees)
- understand the influence and effects of pathogens on honey bee foraging and pollination activities, so that the quality of pollination provided by honey bees can be maintained
- establish practices to maintain, or increase the level of free pollination from wild insects
- develop systems for managing and using alternative pollinators (such as stingless, blue banded and leaf-cutter bees) for specialised production environments
- select stronger self-pollinating traits in relevant crop cultivars or species
- investigate artificial or mechanical pollination in relevant crop cultivars or species.

- Keeping managed honey bees healthy

Effective varroa mite management strategies are only possible with a clear understanding of the biology and pathology of the varroa mite – honey bee interaction. Recent publication of the honey bee genome and work underway on the varroa mite genome will help advance this research.

Australian scientists are well placed to contribute to the global research program that is underway.

By participating, Australian scientists will be better placed to rapidly transfer the benefits of this research to the Australian honey bee industry, and pollination-dependent plant industries.

Bee pathology research also needs to focus on the cluster of viruses and other pathogens that kill bees weakened by the varroa mite. The pattern of secondary infection is complex and likely to differ from place to place. Research and development is needed to understand the role of these pathogens in bee mortality, and the scope for directly reducing the impact of secondary infection. Research and development is needed to:

- improve our understanding of the genetic basis of honey bee immunity (tolerance/resistance) to pests and diseases which will allow more effective selection of high-quality breeding lines
 - identify novel ways to subvert the varroa mite's reproductive cycle
 - develop rapid field tests for key honey bee pathogens
 - develop bio-control agents for the varroa mite
 - understand key mechanisms and variations in immune functionality in managed and wild honey bees
 - identify immune functionalities that help combat major pests and diseases.
- Improving the efficiency of beekeeping and facilitating expansion of the paid pollination services sector

The varroa mite will significantly change the cost structure of Australian beekeeping businesses. Labour costs will rise as beekeepers are forced to monitor hives more intensely and apply appropriate treatments or cultural practices. Research and development is needed to:

- develop equipment or practices to increase the labour efficiency of beekeeping businesses
 - educate Australian beekeepers on the varroa mite and how to adopt effective mite monitoring into their apiary practices
 - assess, compile and report approaches developed by overseas beekeepers to reduce the labour costs from managing with the varroa mite.
- Refining surveillance and monitoring systems

Australia enjoys freedom from a wide range of exotic bees and bee pests. We have a range of awareness, surveillance and quarantine measures in place across the biosecurity continuum (pre-border, border and post-border) to maintain this freedom. However, we need to further refine

our surveillance and monitoring systems in order to detect a varroa mite incursion early enough to have a chance of eradication.

Research and development is needed to:

- determine the sensitivity of sentinel hives to detect exotic honey bee mites such as varroa
- refine remote surveillance catch-box technology and placement at high risk ports
- conduct surveillance auditing, training and simulation workshops
- develop tracking systems (similar to the National Livestock Identification Scheme) that could be used in a varroa mite outbreak
- develop tools to measure the location and spread of an incursion.

Implementation—building on existing research and funding

Scientists based at Australian universities, the CSIRO and state government agencies, in partnership with funding agencies such as the Rural Industries Research and Development Corporation and the Australian Research Council, have built a body of work on the biology and genetics of the European honey bee and the varroa mite, and the pollination of crops in Australia. Substantial related research is also underway in other countries. This provides a solid foundation for funding future work as outlined in this statement.

Globally, only the Australian honey bee industry remains free of varroa.

Sixty-five per cent of horticultural and agricultural crops grown in Australia respond to honey bee pollination.

Source: <https://www.agriculture.gov.au/biosecurity-trade/pests-diseases-weeds/bees/honey-bee-pollination-continuity-strategy/statement-research-development>